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Food knowledge graph for a health tracking system

Abstract

A method of operating a health tracking system is disclosed. The method comprises: storing a food knowledge graph having a plurality of labels describing consumable items and a plurality of relationships between pairs of labels, some of the labels being generic names for consumable items; receiving a data record having a descriptive string regarding a consumable item from a first health tracking device; matching the descriptive string to at least one label in the plurality of labels; and updating one or more information fields of the data record to associate the data record with the at least one label to which the descriptive string was matched. In some embodiments, the method further includes receiving a request for data records from a health tracking device and matching the request to the plurality of labels to provide an improved response to the request for data records.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This document is a continuation of U.S. patent application Ser. No. 16/417,291, filed may 20, 2019, now U.S. Pat. No. 11,887,719,

which claims priority to U.S. Provisional Patent Application Ser. No. 62/674,170, filed May 21, 2018, the entire contents of which are incorporated by reference herein.

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FIELD

(2) The methods and systems disclosed in this document relate health tracking systems having a food database and, more particularly, to a food knowledge graph for supplementing the food database.

BACKGROUND

(3) In recent years, health and fitness tracking applications that track food consumption have become very popular. Food consumption is important to a healthy lifestyle and a person's diet is well known to be related to various health conditions, such as diabetes and obesity to name a few. These applications enable users to gain insights that help them make smarter choices and create healthier habits. In order to track food consumption, such health and fitness tracking applications typically utilize a large crowdsourced database of foods and beverages that can be logged by users. The largest databases include records having nutritional information for millions of foods and beverages. However, these records generally include very minimal information about the food, which is often limited to nothing more than a basic user-generated text description and basic nutritional content information. Accordingly, user interaction with the database is often limited to basic text searching of the database. As a result, it is often cumbersome for users to find foods and beverages in the database. Accordingly, it would be advantageous to provide users with health tracking systems that includes records having more robust information about foods, thereby enabling more informative, efficient, and diverse user interactions with the database.

SUMMARY

(4) In accordance with one exemplary embodiment of the disclosures, a method of operating a health tracking system is disclosed. The method comprises: storing, with a processor, in a database, a plurality of labels describing consumable items and a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items; receiving, with the processor, a data record comprising at least one descriptive string regarding a first consumable item from a first health tracking device; matching, with the processor, the at least one descriptive string to at least one first label in the plurality of labels; updating, with the processor, at least one information field of the data record to associate the data record with the at least one first label to which the at least one descriptive string was matched; receiving, with the processor, a request for data records from a second health tracking device; matching, with the processor, the request to the at least one first label; and providing, with the processor, the data record having the updated at least one information field to the second health tracking device in response to request being matched to the at least one first label.

(5) Pursuant to another exemplary embodiment of the disclosures, a health tracking system is disclosed. The health tracking system comprises: an interface configured to communicate with a health tracking device of a user; a database configured to store (i) a plurality of labels describing consumable items and a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items, and (ii) a plurality of consumable records, each consumable record including at least a descriptive string regarding a consumable item and each consumable record being associated with at least one label in the plurality of labels; and a data processing system in communication with the health tracking device via the interface and the database. The data processing system is configured

to: receive a search string from the health tracking device of the user via the interface; match the search string to at least one first label in the plurality of labels; generate a list of consumable records including consumable records in the plurality of consumable records that are associated with the at least one first label to which the search string was matched; and provide the list of consumable records to the health tracking device as search results.

(6) In accordance with yet another exemplary embodiment, a further method of operating a health tracking system is disclosed. The method comprises: storing, with the processor, in a database, (i) a plurality of labels describing consumable items and a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items, and (ii) a plurality of consumable records, each consumable record including at least a descriptive string regarding a consumable item and each consumable record being associated with at least one label in the in the plurality of labels; receiving, with the processor, from a user, at least one consumable item selection corresponding to a consumable record in the plurality of consumable records in the database; identifying, with the processor, at least one first label in the plurality of labels that is associated with the consumable record in the plurality of consumable records corresponding to the at least one consumable item selection; and providing, with the processor, a list of consumable records to the health tracking device as recommended consumable items for the user based on the at least one first label.

(7) These and other aspects of the disclosure shall become apparent when considered in light of the disclosure provided herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing aspects and other features of a health tracking system are explained in the following description, taken in connection with the accompanying drawings.

(2) FIG. 1 shows a health tracking system.

(3) FIG. 2 shows a system server or data processing system of the health tracking system.

(4) FIG. 3 shows a smart phone of the health tracking system.

(5) FIG. 4A shows an exemplary partial food knowledge graph illustrating ontological relationships that interconnect consumables names of an ingredient sub-graph.

(6) FIG. 4B shows a further exemplary partial food knowledge graph illustrating component relationships that interconnect consumables names of a dish/recipe sub-graph with an ingredient sub-graph and an allergen sub-graph.

(7) FIG. 4C shows a yet another exemplary partial food knowledge graph illustrating dietary substitute relationships that interconnect consumables names of an ingredient sub-graph.

(8) FIG. 4D shows a yet another exemplary partial food knowledge graph illustrating attribute relationships that interconnect consumables names of a dish/recipe sub-graph with a flavor sub-graph, a cuisine sub-graph, and a dietary restriction sub-graph.

(9) FIG. 4E shows a yet another exemplary partial food knowledge graph illustrating ontological relationships between brand names of a brand name sub-graph and brand relationships between branded consumable names of a dish/recipe sub-graph and brand names of the brand name sub-graph.

(10) FIG. 5 shows a method of operating the health tracking system to generate supplemental information for consumable records using the food knowledge graph.

(11) FIG. 6A shows a food labeler of the health tracking application.

(12) FIG. 6B shows an exemplary matching of a consumable record to the food knowledge graph.

(13) FIG. 6C shows an exemplary updated consumable record which has been updated based on the labels to which the consumable record of FIG. 6B was matched.

(14) FIG. 7 shows a method of operating the health tracking system to provide improved search results using the food knowledge graph.

(15) FIG. 8A shows an exemplary graphical user interface including results of a search was performed only on the basis user-generated descriptions of the consumable records.

(16) FIG. 8B shows an exemplary graphical user interface in which search results are populated using the food knowledge graph.

(17) FIG. 9 shows a method of operating the health tracking system to provide improved recommendations using the food knowledge graph.

(18) FIG. 10A shows an exemplary graphical user interface in which recommendations of similar consumables are provided using the food knowledge graph.

(19) FIG. 10B shows an exemplary graphical user interface in which user-specific recommendations of consumables are provided using the food knowledge graph.

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DETAILED DESCRIPTION

(21) In the following detailed description, reference is made to the accompanying figures which form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense, and the scope of embodiments is defined by the appended claims and their equivalents.

(22) Aspects of the disclosure are disclosed in the accompanying description. Alternate embodiments of the present disclosure and their equivalents may be devised without parting from the spirit or scope of the present disclosure. It should be noted that any discussion herein regarding “one embodiment”, “an embodiment”, “an exemplary embodiment”, and the like indicate that the embodiment described may include a particular feature, structure, or characteristic, and that such particular feature, structure, or characteristic may not necessarily be included in every embodiment. In addition, references to the foregoing do not necessarily comprise a reference to the same embodiment. Finally, irrespective of whether it is explicitly described, one of ordinary skill in the art would readily appreciate that each of the particular features, structures, or characteristics of the given embodiments may be utilized in connection or combination with those of any other embodiment discussed herein.

(23) Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order than the described embodiment. Various additional operations may be performed and/or described operations may be omitted in additional embodiments.

(24) For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B and C).

(25) The terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous.

(26) As used herein, the term “consumable” or “consumable item” refers to foods, beverages, dietary supplements, vitamin supplements, medication, and other items for consumption. As used herein, the term “consumable record” or “consumable item record” refers to a database record that relates to a particular consumable. Each consumable record comprises a plurality of data fields that relate to a particular consumable item. In some embodiments, each consumable record includes a description field that includes data, such as a text string, that identifies or describes the particular

consumable. In some embodiments, each consumable record includes a brand field that includes data, such as a text string, that identifies or describes the brand of the particular consumable. In some embodiments, each consumable record includes an ingredients field that includes data, such as one or more text strings, that list ingredients for a particular consumable. In some embodiments, each consumable record includes fields for caloric content, macronutrients, micronutrients, serving size, and other nutrition and health information.

(27) Health Tracking System

(28) With reference to FIG. 1, an exemplary embodiment of a health tracking system **100** that utilizes a food knowledge graph to provide structured and detailed knowledge about consumables is shown. In the illustrated embodiment, the health tracking system **100** includes a plurality of health tracking devices **110** in communication with a system server **200** or other data processing system over a network **120** such as, e.g. the Internet.

(29) The server **200** comprises a computerized device or data processing system configured to run one or more software applications on a processor thereof (e.g. the network-side health tracking program **218**). The server **200** of the present embodiment is further configured to receive a plurality of consumable records which include item descriptions, as well as caloric and nutritional contents of a respective plurality of consumable items which are entered at the health tracking devices **110**, other consumer devices, and/or provided from one or more manufacturing or distributing entities. The consumable records are stored at a storage apparatus or memory of the server **200** (e.g., consumable records **224**).

(30) The storage apparatus or memory is configured to store instructions including a network-side health tracking program **218** (which may also be referred to herein as the “health tracking application”), as well as a database **220** accessible by at least the health tracking program **218**. The database **220** includes user data **222**, consumable records **224**, operational records **226**, and graphics **228**. Alternatively, the server **200** may be in communication with a separate storage entity (not shown) for storage thereof.

(31) As will be discussed in further detail elsewhere herein, the server **200** utilizes a food knowledge graph **230**, which is also included in the database **220**, to provide structured and detailed knowledge about consumables to which the consumable records **224** correspond. In one embodiment, the food knowledge graph **230** is used to generate additional metadata which is included in the consumable records **224**, in addition to the basic data entered by a user upon creation each consumable record. In one embodiment, the food knowledge graph **230** is used to provide more relevant search results when a user searches the consumable records **224**. In one embodiment, the food knowledge graph **230** is used to provide more relevant recommendations of consumable records **224** to the user.

(32) The health tracking devices **110** (which may also be referred to herein as “health and fitness tracking devices”) comprise any number of computerized apparatus, which include a user interface, such as e.g., a smartphone **110A**, laptop computer **110B**, a tablet computer, a smart watch, a desktop computer **110C**, or other such device. In at least one embodiment, the user interface may comprise an LCD touch screen or the like, a mouse or other pointing device, a keyboard or other keypad, speakers, and a microphone, as will be recognized by those of ordinary skill in the art. The user interface provides the user with any of various health, fitness and activity related data such as food and nutritional consumption, calorie expenditure, sleep metrics, weight, body fat, heart rate, distance travelled, steps taken, etc. In order to connect to the network **120**, the health tracking devices **110** are generally configured to utilize any of various wired or wireless communications components, infrastructures and systems, such as cell towers **115** of a mobile telephony network, wireless routers **125**, Bluetooth®, near field communication (NFC), or physical cables. Health tracking devices **110** may use data collected from sensors associated to or in communication with the health tracking device **110**, such as heart rate monitors, step counters, stair counters, global positioning system (“GPS”) tracking devices, as well as various other motion tracking and

biometric monitoring devices. Alternatively, or in addition, a user may manually enter health related data. Such sensors allow the user to easily track and automatically log activity and/or consumption information with the health tracking device. In addition, the health tracking device **110** may include one or more cameras configured to obtain health parameter data including e.g., capture images of a user's performance of an activity and/or capture images of consumed items or descriptions thereof (including barcodes or other machine readable identifiers).

(33) The health tracking devices **110** are configured to communicate with the system server **200** in order to enable: accessing and searching of the consumable records **224** stored thereat, display of the consumable records, provide additional records, and/or enable the user to select individual ones of the displayed consumable records for the purposes of caloric and nutritional logging. In one embodiment, foregoing functions are performed via execution of one or more software applications at the server **200** (i.e., server or network-side applications) in communication with one or more complementary software applications at the health tracking devices **110** (i.e., client-side applications). For example, the health tracking program **218**, running on the processor (of the server **200**) may be utilized to accomplish the foregoing, as explained in further detail below. A client-side software application for performing various functions necessary for the herein disclosed concepts may also be utilized (see health tracking application **316** of FIG. 3, discussed below).

(34) System Server

(35) With reference now to FIG. 2, a block diagram of an exemplary embodiment of the system server **200** of FIG. 1 is shown. It is appreciated that the embodiment of the system server **200** shown in FIG. 2 is only one exemplary embodiment of a system server **200**. As such, the exemplary embodiment of the system server **200** of FIG. 2 is merely representative of any of various manners or configurations of system servers or other data processing systems that are operative in the manner set forth herein.

(36) The system server **200** of FIG. 2 is typically provided in a housing, cabinet or the like **202** that is configured in a typical manner for a server or related computing device. In one embodiment, the system server **200** includes processing circuitry/logic **204**, memory **206**, a power module **208**, a user interface **210**, a network communications module **212**, and a wireless transceiver **214**.

(37) The processing circuitry/logic **204** is operative, configured and/or adapted to operate the system server **200** including the features, functionality, characteristics and/or the like as described herein. To this end, the processing circuitry/logic **204** is operably connected to the memory **206**, the power module **208**, the user interface **210**, the network communications module **212**, and the wireless transceiver **214**. The memory **206** may be of any type of device capable of storing information accessible by the processor, such as a memory card, ROM, RAM, write-capable memories, read-only memories, hard drives, discs, flash memory, or any of various other computer-readable medium serving as data storage devices as will be recognized by those of ordinary skill in the art. The memory **206** is configured to store instructions including a network-side health tracking application **218** for execution by the processing circuitry/logic **204**, as well as a database **220** for use by at least the health tracking program **218**. The database **220** includes user data **222**, consumable records **224**, operational records **226**, graphics **228**, and the food knowledge graph **230**. As discussed in greater detail below, the health tracking application **218** includes instructions for utilizing the food knowledge graph **230** to provide more informative, efficient, and diverse user interactions with consumable records **224** in the database **220**.

(38) With continued reference to FIG. 2, the power module **208** of the system server **200** is operative, adapted and/or configured to supply appropriate electricity to the system server **200** (i.e., including the various components of the system server **200**). The power module **208** may operate on standard **120** volt AC electricity, but may alternatively operate on other AC voltages or include DC power supplied by a battery or batteries.

(39) The network communication module **212** of the system server **200** provides an interface that allows for communication with any of various devices using various means. In particular, the

network communications module **212** includes a local area network port that allows for communication with any of various local computers housed in the same or nearby facility. In some embodiments, the network communications module **212** further includes a wide area network port that allows for communications with remote computers over the Internet (e.g., network **120** of FIG. **1**). Alternatively, the system server **200** communicates with the network **120** via a modem and/or router of the local area network. In one embodiment, the network communications module is equipped with a wireless transceiver **214** or other wireless communications device. Accordingly, it will be appreciated that communications with the system server **200** may occur via wired communications or via the wireless communications. Communications may be accomplished using any of various known communications protocols. In the embodiment of FIG. **2**, the wireless transceiver **214** may be a Wi-Fi transceiver, but it will be recognized that the wireless transceiver may alternatively use a different communications protocol.

(40) The system server **200** may be accessed locally by an authorized user (i.e., an administrator or operator). To facilitate local access, the system server **200** includes an interactive user interface **210**. Via the user interface **210**, an operator may access the instructions, including the health tracking application **218**, and may collect data from and store data to the memory **206**. In at least one embodiment, the user interface **210** may suitably include an LCD touch screen or the like, a mouse or other pointing device, a keyboard or other keypad, speakers, and a microphone, as will be recognized by those of ordinary skill in the art. Accordingly, the user interface **210** is configured to provide an administrator or other authorized user with access to the memory **206** and allow the authorized user to amend, manipulate and display information contained within the memory.

(41) As mentioned above, the memory **206** includes various programs and other instructions that may be executed by the processor circuitry/logic **204**. In particular, the memory **206** of the system server **200** of FIG. **2** includes the health tracking program **218** (which may also be referred to herein as a “health tracking application”). The health tracking program **218** is configured to cause the system server **200** to enable a user to obtain nutritional data related to any of various consumables. Execution of the health tracking application **218** by the processor circuitry/logic **204** results in signals being sent to and received from the user interface **210** and the communications module **212** (for further delivery to a user device such as a health tracking device **110**), in order to allow the user receive and update various aspects of the consumable records **224**. The network-side health tracking application **218** is configured to provide various graphical views and screen arrangements to be displayed to a user on a health tracking device **110**.

(42) The user data **222** includes at least user profiles **232** and corresponding consumable logs **234**. The user profiles **232** include a profile data for each user of the health tracking system **100**. Each user profile includes demographic information for the users such as name, age, gender, height, weight, performance level (e.g., beginner, intermediate, professional, etc.) and/or other information for the user. In at least one embodiment, the consumable logs **234** include a consumable diary/log for each user (which may also be referred to herein as a “food diary”). The consumable diary/log allows the user to track consumables that are consumed by the user over a period of days and any nutritional data associated with the food consumed. For example, the consumable diary/log may allow the user to enter particular consumable that is consumed by the user and keep track of the associated calories, macronutrients, micronutrients, sugar, fiber, and/or any of various other nutritional data associated with the consumables entered by the user in the consumable diary/log. In some embodiments, the user data **222** further includes various activity and fitness data collected by sensors (not shown) associated with the health tracking devices **110**.

(43) In an alternative embodiment, the foregoing profile data may be stored at a storage entity separate from yet in communication with the server **200**. For example, a centralized server may be provided which is configured to store all data relating to an individual user in one storage area (including workout data, nutrition/consumption data, profile data, etc.).

(44) A plurality of consumable records **224** is stored in the database **220**. As discussed above, the

term “consumable record” refers to a database record that relates to a particular consumable item. In at least one embodiment, each consumable record comprises a plurality of data fields that relate to a particular consumable item. In the disclosed embodiment, each of the consumable records includes a number of fields including, for example, a name for the consumable item, summary information about the consumable item, and detailed nutritional information about the consumable item. Detailed nutritional information about a consumable item may include one or more of: serving size, calories, nutrients, ingredients, or any other nutritional information about the item. For example, the detailed nutritional information may include information that may be provided on USDA food labels or state-regulated food labels (e.g., vitamin and mineral content, fat content, cholesterol content, protein content, sugar content, carbohydrate content, fiber content, organic contents, etc.). The summary information about the consumable may include some subset of the more detailed information about the consumable. For example, the summary information about the consumable may only include serving size and calorie information. The various fields of each consumable record may be populated by data from any user or third party data providers. Many, if not all, of consumable records **224** are created by users of the health tracking system **100** and/or have fields that are editable by users, without the need for special authorization or privileges. However, it will be recognized that in at least some embodiments, consumable records **224** may have been entered by any of various sources including an administrator or operator of the health tracking system **100**, commercial food providers (e.g., food distributors, restaurant owners, etc.), and/or users of the health tracking system **100**. In addition, certain information may be stored in a machine readable code (such as a bar code or QR code) which is captured via a camera or other scanner at the user device **110**.

(45) The operational records **226** include current and historical data stored by the system server **200** in association with operation of the system server **200**, execution of the health tracking application **218**, and/or manipulation of the database **220** within the memory **206**. For example, the operational records **226** may include information concerning amendments made to any of various consumable records **224**. The operational records **226** may also include other information related to the control and operation of the system server **200**, including statistical, logging, licensing, and historical information.

(46) In one embodiment, graphical views **228** are provided at the server **200** which are pushed to the health tracking device **110** for display thereat of various screen arrangements.

(47) While the system server **200** has been explained in the foregoing embodiment as housing the health tracking program **218** and the various records and databases in the memory **206**, it will be recognized that in other embodiments these components may be retained in other one or more remote locations in communication with the health tracking system **100**. For example, in at least one embodiment, the consumable records **224** may comprise data retained by a database separate from the system server **200**. Alternatively, the consumable records **224** or certain fields of the consumable records **224** are received from a third party database. In such embodiments, the health tracking application may utilize any number of application programming interfaces (APIs) to access the data in the third party databases and incorporate such information for use in the health tracking application **218**, without local storage thereof. Accordingly, it will be recognized that the description of the system server **200** of FIG. 2 is but one exemplary embodiment of a data processing system that may be utilized by the health tracking system **100**.

(48) A computer program product implementing an embodiment disclosed herein may therefore comprise one or more computer-readable storage media storing computer instructions executable by a processor to provide an embodiment of a system or perform an embodiment of a method disclosed herein. Computer instructions (e.g., the health tracking application **218**) may be provided by lines of code in any of various languages as will be recognized by those of ordinary skill in the art. A “non-transitory computer-readable medium” may be any type of data storage medium that may store computer instructions, including, but not limited to a memory card, ROM, RAM, write-

capable memories, read-only memories, hard drives, discs, flash memory, or any of various other computer-readable medium.

(49) Health Tracking Devices

(50) With reference again to FIG. 1, the health tracking devices **110** may be provided in any of various forms. Examples of a health tracking devices **110** configured for use with the health tracking system **100** include a smartphone **110A**, a laptop computer **110B**, and a desktop computer **110C**, as shown in FIG. 1, as well as various other electronic devices. Accordingly, it will be recognized that the health tracking devices **110** may comprise portable electronic devices such as the smartphone **110A** or the laptop computer **110B**, or stationary electronic devices such as the desktop computer **110C**. Other examples of health tracking devices include, handheld or tablet computers, smart watches, portable media players, other wearable devices, or any of various other health tracking devices configured to receive entry of consumables (not shown).

(51) In one embodiment, data entered at one device **110** may be provided to other ones of the user's devices **110**. For example, data entered at the smart phone **110A** may be provided to the desktop computer **110C** and/or the laptop computer **110B** for storage thereat. Alternatively or in addition, the data may be stored at a single network storage apparatus (not shown) having a dedicated portion of storage for records relating to the user and accessible by all of the user's devices **110**.

(52) With reference now to FIG. 3, in at least one embodiment the health tracking device **110** is provided in the form of a smartphone **110A**. The smartphone **110A** includes a display screen **302**, an input/output (I/O) interface **304**, a processor **308**, a memory **310**, and one or more transceivers **312**. The smartphone **110A** also includes a protective outer shell or housing **314** designed to retain and protect the electronic components positioned within the housing **314**. The smartphone **110A** also includes a battery (not shown) configured to power the display screen **302**, processor **308**, transceivers **312** and various other the electronic components within the smartphone **110A**.

(53) The display screen **302** of the smartphone **110A** may be an LED screen or any of various other screens appropriate for the personal electronic device. The I/O interface **304** of the smartphone **110A** includes software and hardware configured to facilitate communications with the user. The I/O interface **304** is in communication with the display screen **302** and is configured to visually display graphics, text, and other data to the user via the display screen **302**. As will be recognized by those of ordinary skill in the art, the components of the health tracking device **110** may vary depending on the type of display device used. Alternative health tracking devices, such as the laptop **110B** and the desktop **110C**, may include much of the same functionality and components as the smartphone **110A** shown in FIG. 3, but may not include all the same functionality or components and/or may include others not listed.

(54) The processor **308** of the smartphone **110A** may be any of various processors as will be recognized by those of ordinary skill in the art. The processor **308** is in communication with the I/O interface **304**, the memory **310**, and the transceivers **312**, and is configured to deliver data to and receive data from each of these components. The memory **310** is configured to store information, including data and instructions for execution by the processor **308**. It will be recognized by those of ordinary skill in the art that a "processor" includes any hardware system, hardware mechanism or hardware component that processes data, signals or other information. A processor may include a system with a central processing unit, multiple processing units, dedicated circuitry for achieving functionality, or other systems.

(55) The transceivers **312** may be any of various devices configured for communication with other electronic devices, including the ability to send communication signals and receive communication signals. The transceivers **312** may include different types of transceivers configured to communicate with different networks and systems. Such transceivers are well known and will be recognized by those of ordinary skill in the art.

(56) In some embodiments, the transceivers **312** include at least one transceiver configured to allow the smartphone **110A** to perform wireless communications with the cell towers **115** of the wireless

telephony network, as will be recognized by those of ordinary skill in the art. The wireless telephony network may comprise any of several known or future network types. For example, the wireless telephony network may comprise commonly used cellular phone networks using CDMA, GSM or FDMA communication schemes, as well as various other current or future wireless telecommunications arrangements. In some embodiments, the transceivers **312** include at least one transceiver configured to allow the smartphone **110A** to communicate with any of various local area networks using Wi-Fi, Bluetooth® or any of various other communications schemes.

(57) In some embodiments, the memory **310** includes program instructions for a graphical user interface configured to provide a client-side health tracking application **316**. The memory **310** may further be configured to store certain user data **318**, such as e.g., user gender, height, weight, user identifier, password, etc. Additionally, health related data (e.g., data collected from one or more sensors and/or manually entered) may be stored. The processor **308** is configured to read the program instructions from the memory **310** and execute the program instructions to provide the health tracking application **316** to the user so for the purpose of performing health and fitness related tasks for the user, including displaying, modifying, and analyzing the user data **318**.

(58) In at least one embodiment, the user data **318** includes a plurality of consumable records which serves as a log of consumables that have been consumed by the user for the purpose of caloric and nutritional tracking. That is to say, the client-side health tracking application **316** is configured to display consumable records and enable the user to select consumable records (from a plurality of records accessed via the network **120**), those items that correspond to consumables that he or she has consumed are stored at the client-side for the purpose of logging the consumables in this embodiment. In another alternative, such log may be stored remote from the device and/or only kept at the device for a transitory period.

(59) The memory **310** that retains the data and instructions may be of any type of device capable of storing information accessible by the processor, such as a memory card, ROM, RAM, write-capable memories, read-only memories, hard drives, discs, flash memory, or any of various other computer-readable medium serving as data storage devices as will be recognized by those of ordinary skill in the art. Portions of the system and methods described herein may be implemented in suitable software code that may reside within the memory as software or firmware. Alternatively, or in addition, the software (such as e.g., the client side health tracking program **316**) may be downloaded from a network location, such as via the Internet.

(60) Food Knowledge Graph

(61) As discussed above, the database **220** includes a food knowledge graph **230**, which is utilized by the health tracking application **218** of the server **200** to provide structured and detailed knowledge about consumables to which the consumable records **224** correspond. The food knowledge graph **230** improves upon the health tracking system **100** by enabling more informative, efficient, and diverse user interactions with the consumable records **224** in the database **220**.

(62) As used herein, the phrase “food knowledge graph” or “food graph” refers to a dataset including descriptive labels that describe various aspects of one or more consumables, and defined relationships between two or more of the descriptive labels. The descriptive labels may include any label that provides some description or information regarding a consumable. In some embodiments, a food knowledge graph may take the form of one or more lists or tables that define a data tree or data web in which the descriptive labels comprise the nodes of the tree or web and the relationships comprise the connections between the nodes of the tree or web. In one embodiment, a reduced number of self-referential lists or tables, possible a single self-referential list or table, define all of the labels and relationships that form the data tree or data web. In some embodiments, the labels and/or relationships are stored as a list or table of triplets formed as, for example, [label_ID1, relationship_type, label_ID2] where label_ID1 is a first descriptive label or reference thereto, label_ID2 is a second descriptive label or reference thereto, and relationship_type indicates the nature of the relationship. However, the food knowledge graph can take any other suitable form.

(63) Some exemplary descriptive labels that might be included in a food knowledge graph are generic consumable names (e.g., apple, pizza, sandwich, cola, chocolate chip cookie), brand names (e.g., Kraft®, McDonald's®, Pepsi®), categories of consumables (e.g., fruits, grains, spices, condiments), categories of brand names (e.g. restaurant, fast food, grocery store, national food manufacturer, regional/local food manufacturer), allergens (e.g., nuts, gluten, dairy), flavors (e.g., bitter, salty, umami/savory, sour, spicy, and sweet), types of cuisine (e.g., Mexican, Chinese, Italian), dietary restriction compliances (e.g., paleo, vegan, vegetarian, kosher, halal), and other miscellaneous useful descriptive labels (e.g., sugar-free, low-carb, organic, cage-free, non-GMO).

(64) FIG. 4A shows an exemplary partial food knowledge graph illustrating ontological relationships that interconnect consumables names of an ingredient sub-graph **400**. Particularly, one exemplary relationship that might be defined in a food knowledge graph is an ontological relationship (“is a” and/or “has the category”) indicating that a first consumable name is a subclass of a second consumable name, or is within a particular category of consumables. For example, a pepperoni pizza “is a” subclass of pizza and an apple “has the category” fruit. Ontological relationships are not, however, limited to consumable names. As will be seen below, any type of descriptive label can have similar ontological relationships.

(65) The ingredient sub-graph **400** includes a plurality of generic consumable name labels (indicated by a rectangle) and a plurality of categories of consumables (indicated by an oval). As used herein, “ingredients” (which may also be referred to herein as “simple foods” or “simple consumables”) refer to consumables and/or consumable names that are not decomposed into further subcomponents in the food knowledge graph.

(66) The ingredient sub-graph **400** includes a plurality of ontological relationships which indicate that a consumable name is a subclass of another consumable name (indicated by a solid arrow). Particularly, as illustrated in FIG. 4A, the food knowledge graph defines that “green kale” and “red kale” are subclasses of “kale,” that “olive oil” and “canola oil” are subclasses of “vegetable oil,” that “lard” is a subclass of “animal fat,” and that “2% milk” and “skim milk” are subclasses of “milk.” Similarly, the ingredient sub-graph **400** further includes a plurality of ontological relationships which indicate that a category of consumable names is a subclass of another category of consumable name (also indicated by a solid arrow). Particularly, as illustrated in FIG. 4A, the food knowledge graph defines that “allium” is a subclass of “vegetable,” that “vegetable” is a subclass of “plant source,” that “dairy” is a subclass of “animal source,” and “plant source,” “oil & fat,” and that “animal source” are subclasses of “ingredient.”

(67) Furthermore, the ingredient sub-graph **400** includes a plurality of ontological relationships which indicate that a consumable name is within a particular category of consumable names (indicated by a dashed arrow). Particularly, as illustrated in FIG. 4A, the food knowledge graph defines that “onion” and “garlic” have the category “allium,” that “kale” and “carrot” have the category “vegetable,” that “vegetable oil” has the categories “plant source” and “oil & fat,” that “fish oil” and “animal fat” have the categories “animal source” and “oil & fat,” that “egg” has the category “animal source,” and that “milk,” and “butter” have the category “dairy.”

(68) In the embodiment shown, a particular consumable name is only a direct subclass of a single other consumable name. Furthermore, consumable names are generally only related to the narrowest appropriate category or class. For example “onion” is related to the narrower “allium” category, rather than to the broader “vegetable” or “plant source” categories. In some instances, however, consumable names are related to multiple categories when one category is not strictly a subclass of the other. For example, “vegetable oil” is within both the “plant source” and “oil & fat” categories because “oil & fat” is not strictly a subclass of “plant source” (i.e. consumables in “oil & fat” category could be a “plant source” or an “animal source”). Depending on the design of the sub-graph, different suitable rules for ontological relationships may be applied.

(69) FIG. 4B shows a further exemplary partial food knowledge graph illustrating component relationships that interconnect consumables names of a dish/recipe sub-graph **402** with an

ingredient sub-graph **404** and an allergen sub-graph **406**. Particularly, another exemplary relationship that might be defined in a food knowledge graph is a component relationship (“contains”) indicating that a first consumable name contains a second consumable name as a component and/or ingredient. For example, a pizza “contains” cheese, lemonade “contains” a sweetener, a cinnamon roll “contains” cinnamon. One variant of the component relationship that might be defined in a food knowledge graph is an allergen relationship (“has the allergen”) indicating that a first consumable name contains a particular allergen. For example, a peanut butter cup “has the allergen” nuts, chocolate milk “has the allergen” dairy, and a breadstick “has the allergen” gluten.

(70) The dish/recipe sub-graph **402** and ingredient sub-graph **404** each include a plurality of generic consumable name labels (indicated by a rectangle) and a plurality of categories of consumables (indicated by an oval). As used herein, “dish” and “recipe” (which may also be referred to herein as “compound foods”) each refer to consumables and/or consumable names that are decomposed into further subcomponents in the food knowledge graph. In some embodiments, “dish” and “recipe” correspond to different types of consumable records in the database **220**, but may be treated as equivalent with respect to the food knowledge graph **230**. In other embodiments, separate sub-graphs may be used for “dishes” and “recipes.” The ingredient sub-graph **404** is essentially similar to the ingredient sub-graph **400** of FIG. 4A and not described in complete detail. The dish/recipe sub-graph **402** defines ontological relationships between a plurality of dishes and/or recipes, and is essentially analogous to the ingredient sub-graph **400** of FIG. 4A and not described in complete detail. Finally, the allergen sub-graph **406** includes a plurality of allergen labels (indicated by a triangle), and although not shown, may define similar ontological relationships as the aforementioned sub-graphs **400**, **402**, and **404**.

(71) The sub-graphs **402**, **404**, and **406** include a plurality of component relationships which indicate that a consumable name includes a particular consumable name (indicated by a dotted arrow). Particularly, as illustrated in FIG. 4B, the food knowledge graph defines that a “peanut butter chocolate chip cookie” contains “cookie dough,” “peanut butter,” and a “chocolate chip.” As can be seen dishes and recipes names, a “peanut butter chocolate chip cookie,” may contain simple ingredients (e.g. chocolate chips and peanut butter) as well as other compound dishes or recipes (e.g. cookie dough). Particularly, the food knowledge graph defines that the “cookie dough” component itself contains the ingredients “egg,” “milk,” “butter,” “flour,” “vanilla,” “baking powder,” and “sugar.”

(72) Furthermore, the sub-graphs **402**, **404**, and **406** include a plurality of allergen relationships which indicate that a consumable name includes a particular allergen (also indicated by a dotted arrow). Particularly, as illustrated in FIG. 4B, the food knowledge graph defines that a “peanut butter chocolate chip cookie” has the allergens “gluten” and “nut.” Additionally, the food knowledge graph defines that “peanut butter” has the allergen “nut” and that “flour” contains “gluten.” In this way, the food knowledge graph not only indicates what allergens are contained in a “peanut butter chocolate chip cookie,” but also provides sufficient information to eliminate the allergen from a recipe. Particularly, to eliminate the “nut” allergen from a recipe for a “peanut butter chocolate chip cookie,” the “peanut butter” ingredient must be omitted or substituted. Similarly, to eliminate the “gluten” allergen from a recipe for a “peanut butter chocolate chip cookie,” the “flour” ingredient must be omitted or substituted (such as with a gluten-free flour substitute). As will be discussed in further detail below, the food knowledge graph also provides information regarding dietary substitutes of individual consumable names.

(73) FIG. 4C shows a yet another exemplary partial food knowledge graph illustrating dietary substitute relationships that interconnect consumables names of an ingredient sub-graph **408**. Particularly, another exemplary relationship that might be defined in a food knowledge graph is a substitute relationship (“has the substitute”) indicating that a first consumable name has as a dietary substitute a second consumable name. For example, ketchup “has the substitute” mustard, cream

“has the substitute” milk, and sugar “has the substitute” stevia extract. In some embodiments, the food knowledge graph may define multiple types of substitutions. In one embodiment, the food knowledge graph separately defines substitutes that are sufficiently similar to substitute in a recipe (e.g., butter “has the substitute” margarine). In some embodiments, the food knowledge graph separately defines substitutes that are similar in flavor (e.g., grape jelly “has the substitute” strawberry jelly). In some embodiments, the food knowledge graph separately defines substitutes that are sufficiently similar in function (e.g., ketchup “has the substitute” mayonnaise).

(74) The ingredient sub-graph **408** includes a plurality of generic consumable name labels (indicated by a rectangle) and a plurality of categories of consumables (indicated by an oval). The ingredient sub-graph **408** is essentially similar to the ingredient sub-graph **400** of FIG. 4A and not described in complete detail. The ingredient sub-graph **408** includes a plurality of dietary substitute relationships which indicate that a consumable name has as a dietary substitute another consumable name (indicated by a double sided dashed arrow). Particularly, as illustrated in FIG. 4C, the food knowledge graph defines that vegetable oil “has the substitute” coconut oil, and visa versa. Additionally, the food knowledge graph defines that olive oil “has the substitute” canola oil, and visa versa. Furthermore, since olive oil and canola oil are subclasses of vegetable oil, it is known that both olive oil and canola oil also “have the substitute” coconut oil. Similarly, the food knowledge graph defines that rice, quinoa, and couscous are each substitutes for one another. Additionally, the food knowledge graph defines that white rice “has the substitute” brown rice, and visa versa. Furthermore, since white rice and brown rice are subclasses of rice, it is known that both white rice and brown rice also “have the substitutes” quinoa and couscous, and visa versa. Finally, the food knowledge graph defines that milk, soy milk, and almond milk are each substitutes for one another. Additionally, the food knowledge graph defines that 2% milk “has the substitute” skim milk, and visa versa. Furthermore, since 2% milk and skim milk are subclasses of milk, it is known that both 2% milk and skim milk also “have the substitutes” soy milk and almond milk, and visa versa.

(75) In the embodiment shown, when a particular consumable name has as a substitute a class of consumables and its respective subclasses of consumables, it is only directly related to the broader class of consumables. For example, coconut oil only has a direct dietary substitute relationship with vegetable oil, and it is simply implied that all of the subclasses of vegetable oil (i.e. olive oil and canola oil, as shown) are also substitutes. However, in other embodiments, a particular consumable name might also have a directly defined dietary substitute relationship with a broader class of consumables and every subclass thereof.

(76) FIG. 4D shows a yet another exemplary partial food knowledge graph illustrating attribute relationships that interconnect consumables names of a dish/recipe sub-graph **410** with a flavor sub-graph **412**, a cuisine sub-graph **414**, and a dietary restriction sub-graph **416**. Particularly, one exemplary attribute relationship that might be defined in a food knowledge graph is a flavor relationship (“has the flavor”) indicating that a first consumable name has as a particular flavor. For example, a pretzel “has the flavor” salty, cake “has the flavor” sweet, and a lemon “has the flavor” sour. Another exemplary attribute relationship that might be defined in a food knowledge graph is a cuisine relationship (“is from the cuisine”) indicating that a first consumable name is from a particular cuisine. For example, spaghetti “is from the cuisine” Italian, curry chicken “is from the cuisine” Indian, and spring roll “is from the cuisine” Chinese. Yet another exemplary attribute relationship that might be defined in a food knowledge graph is a dietary restriction relationship (“is compliant with”) indicating that a first consumable name is compliant with a particular dietary restriction. For example, a ceasar salad “is compliant with” vegetarian, tofu “is compliant with” vegan, and an all-beef hot dog “is compliant with” kosher.

(77) The dish/recipe sub-graph **410** includes a plurality of generic consumable name labels (indicated by a rectangle) and a plurality of categories of consumables (indicated by an oval). The dish/recipe sub-graph **410** is essentially similar to the ingredient sub-graph **400** of FIG. 4A and not

described in complete detail. The flavor sub-graph **412** includes a plurality of flavors (indicated by a rhombus). As shown, the flavor sub-graph **412** only includes the fundamental flavors. However, in some embodiments, the flavor sub-graph **412** may also include compound flavors (e.g., buttery, tangy, fruity) with ontological relationships defined therebetween. Additionally, the cuisine sub-graph **414** includes a plurality of cuisines (indicated by a parallelogram) and ontological relationships defined therebetween. Finally, the cuisine sub-graph **414** includes a plurality of dietary restrictions (indicated by a trapezoid) and, although none are shown, may also include ontological relationships defined therebetween.

(78) The sub-graphs **410** and **412** are connected by a plurality of flavor relationships which indicate that a consumable name has a particular flavor (indicated by a dash-dotted arrow). Particularly, as illustrated in FIG. 4D, the food knowledge graph defines that a pepperoni pizza “has the flavors” savory and salty. Similarly, the food knowledge graph defines that an “apple pie “has the flavors” sweet, sour, and savory. Finally, the food knowledge graph defines that sweet & sour sauce “has the flavors” sweet, sour, and salty.

(79) The sub-graphs **410** and **414** are connected by a plurality of cuisine relationships which indicate that a consumable name is from a particular cuisine (also indicated by a dash-dotted arrow). Particularly, as illustrated in FIG. 4D, the food knowledge graph defines that a pepperoni pizza “is from the cuisines” Italian and American. Similarly, the food knowledge graph defines that an “apple pie “is from the cuisine” American. Finally, the food knowledge graph defines that sweet & sour sauce “is from the cuisine” Chinese.

(80) The sub-graphs **410** and **416** are connected by a plurality of dietary restriction relationships which indicate that a consumable name is compliant with a particular dietary restriction (also indicated by a dash-dotted arrow). Particularly, as illustrated in FIG. 4D, the food knowledge graph defines that an “apple pie “is compliant with” kosher and vegetarian dietary restrictions, but not a vegan dietary restriction. Similarly, the food knowledge graph defines that sweet & sour sauce “is compliant with” kosher, vegan, and vegetarian dietary restrictions. As can be seen, the food knowledge graph does not define that a pepperoni pizza “is compliant with” any of the illustrated dietary restrictions.

(81) FIG. 4E shows a yet another exemplary partial food knowledge graph illustrating ontological relationships between brand names of a brand name sub-graph **418** and brand relationships between branded consumable names of a dish/recipe sub-graph **420** and brand names of the brand name sub-graph **418**. Particularly, in some embodiments, the descriptive labels may include popular branded consumable names (e.g., Kraft Easy Mac®, McDonald's McNuggets®), in addition to generic consumable names. In such embodiments, another exemplary relationship that might be defined in a food knowledge graph is a brand relationship (“has the brand” or “sells”) indicating that a branded consumable name has a particular brand name or that a particular brand name sells a particular branded consumable name. For example, Mountain Dew® “has the brand” Pepsi® and McDonald's® “sells” Big Mac®.

(82) The brand name sub-graph **418** includes a plurality of brand names (indicated by a rounded rectangle) and a plurality of categories of brand names (indicated by a pentagon). The dish/recipe sub-graph **420** includes a plurality of generic consumable name labels (indicated by a rectangle), branded consumable name labels (indicated by a rectangle having a beveled corner) and a plurality of categories of consumables (indicated by an oval). The dish/recipe sub-graph **420** is essentially similar to the ingredient sub-graph **400** of FIG. 4A and not described in complete detail.

(83) The sub-graphs **418** and **420** are connected by a plurality of brand relationships which indicate that a branded consumable name has a particular brand name or that a particular brand name sells a particular branded consumable name (indicated by a double-sided dotted arrow). Particularly, as illustrated in FIG. 4E, the food knowledge graph defines that Pizza Hut® Pepperoni Pizza “has the brand” Pizza Hut® and, similarly, that Pizza Hut® “sells” Pizza Hut® Pepperoni Pizza. Additionally, the food knowledge graph defines that Kroger® Pepperoni Pizza “has the brand”

Kroger® and, similarly, that Kroger® “sells” Kroger® Pepperoni Pizza. Finally, the food knowledge graph defines that Wendy's® french fries “has the brand” Wendy's® and, similarly, that Wendy's® “sells” Wendy's® french fries.

(84) Furthermore, the ingredient sub-graph **418** includes a plurality of ontological relationships which relate the brand names and the categories of brand names in a manner similar to the ingredient and dish/recipe sub-graphs discussed above. Particularly, as illustrated in FIG. **4E**, the food knowledge graph defines that Wendy's® “has the category” fast food, which itself “is a” subclass of restaurant. Similarly, the food knowledge graph defines that Pizza Hut® “has the category” restaurant. Finally, the food knowledge graph defines that Kroger® “has the category” grocery store.

(85) Methods of Using the Food Knowledge Graph

(86) Various methods and processes for operating a health tracking system are described below. In the description of the methods, statements that a method is performing some task or function refers to a controller or general purpose processor executing programmed instructions stored in non-transitory computer readable storage media operatively connected to the controller or processor to manipulate data or to operate one or more components in the health tracking system **100** to perform the task or function. Particularly, the processor circuitry/logic **204** of the system server **200** and/or the processor **308** of the smartphone **110A** above may be such a controller or processor.

Alternatively, the controller may be implemented with more than one processor and associated circuitry and components, each of which is configured to form one or more tasks or functions described herein. Additionally, the steps of the methods may be performed in any feasible chronological order, regardless of the order shown in the figures or the order in which the steps are described.

(87) FIG. **5** shows a method **500** of operating the health tracking system **100** to generate supplemental information for consumable records using the food knowledge graph **230**. The method **500** improves the functioning of the system server **200** by enabling the processor circuitry/logic **204** to utilize the structured knowledge defined by the food knowledge graph **230** to provide additional information and/or metadata which is stored in or referenced by consumable records, in addition to the user-generated information of the consumable records. Particularly, in some embodiments, many, if not all, of consumable records **224** are created by users of the health tracking system **100** and/or have fields that are editable by users, without the need for special authorization or privileges. However, the user-generated information in the consumable records **224** is often limited to nothing more than a basic user-generated text description and basic nutritional content information. The method **500** supplements this basic user-generated information by labeling the consumable records **224** with additional information such as a standardized generic consumable name, a consumable category, a standardized brand name, a brand name category, dietary substitutes, included ingredients, included allergens, flavors, a type of cuisine, dietary restriction compliance, and other miscellaneous useful descriptive labels.

(88) The method **500** begins with a step of storing a plurality of labels describing consumable items and a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items (block **510**). Particularly, with respect to the embodiments disclosed in detail herein, the processor circuitry/logic **204** of the system server **200** is configured to operate the database **220** to store the food knowledge graph **230**. The food knowledge graph **230** comprises a plurality of descriptive labels describing consumable items and a plurality of relationships between pairs or groups of labels in the plurality of descriptive labels. In one embodiment, the descriptive labels of the food knowledge graph **230** at least include generic consumable names, but may include any other type of descriptive label discussed above with respect to FIGS. **4A-4E** or any other useful descriptive labels not discussed above. Furthermore, the plurality of relationships of the food knowledge graph **230** may include any of the relationship types discussed above with respect to FIGS. **4A-4E**, as well as any other

useful relationship type not discussed above. In some embodiments, the processor circuitry/logic **204** is configured to operate the database **220** to store the food knowledge graph **230** in the form of one or more lists or tables that define a data tree or data web in which the descriptive labels comprise the nodes of the tree or web and the relationships comprise the connections between the nodes of the tree or web (for example, as illustrated in FIGS. 4A-4E). In one embodiment, the processor circuitry/logic **204** is configured to operate the database **220** to store the food knowledge graph **230** in the form of a reduced number of self-referential lists or tables, possible a single self-referential list or table, that define all of the labels and relationships that form the data tree or data web. In some embodiments, the processor circuitry/logic **204** is configured to operate the database **220** to store the labels and/or relationships as a list or table of triplets formed as, for example, [label_ID1, relationship_type, label_ID2] where label_ID1 is a first descriptive label or reference thereto, label_ID2 is a second descriptive label or reference thereto, and relationship_type indicates the nature of the relationship. However, the food knowledge graph can take any other suitable form.

(89) The method **500** continues with a step of receiving a data record comprising at least one descriptive string regarding a first consumable item from a first health tracking device (block **520**). Particularly, the processor circuitry/logic **204** is configured to operate the network communications module **212** to receive a consumable record from a health tracking device **110**, such as the smartphone **110A**. Alternatively, the processor circuitry/logic **204** is configured to read from the database **220**, a consumable record which was previously received from a health tracking device **110**. The received consumable record includes at least one descriptive string that describes a consumable item, which was entered by the user who created the consumable record. In some embodiments, the received consumable record includes both an item description string that describes the consumable item and a brand description string that describes a brand of the consumable item, which were entered by the user who created the consumable record. In the case that the received consumable record is a recipe, the consumable record may further include one or more ingredient description strings. In some embodiments, the received consumable record further includes nutritional content information, which was entered by the user who created the consumable record.

(90) The method **500** continues with a step of matching the at least one descriptive string to at least one first label in the plurality of labels (block **530**). Particularly, the processor circuitry/logic **204** is configured to execute a food labeling process of the health tracking application **218** to match the at least one descriptive string with one or more descriptive labels of the food knowledge graph **230**. FIG. 6A shows a food labeler **600** of the health tracking application **218** of the system server. The food labeler **600** is configured to, when executed by the processor circuitry/logic **204**, match at least one descriptive string **602** with one or more descriptive labels of the food knowledge graph **230** and output a set of matched labels **604**. In one embodiment, the food labeler **600** is configured to, when executed by the processor circuitry/logic **204**, compare the text of a respective descriptive string with the labels and match the respective descriptive string to the most similar label of the food knowledge graph **230** or labels of the food knowledge graph **230** having a threshold level of similarity. In some embodiments, rather than using a rules-based text comparison, the food labeler **600** is configured to, when executed by the processor circuitry/logic **204**, instead use a machine learning model to match the descriptive strings of the received consumable record to labels of the food knowledge graph **230**.

(91) In one embodiment, the food labeler **600** includes three separate labelers, a dish labeler **606**, a brand labeler **608**, and an ingredient labeler **610**. The dish labeler **606** is configured to, when executed by the processor circuitry/logic **204**, match an item description string of the received consumable record with at least one label of the food knowledge graph **230**. Particularly, in one embodiment, the dish labeler **606** is configured to match the item description string of the received consumable record with a most similar generic consumable name label of the food knowledge

graph **230**. Similarly, the brand labeler **608** is configured to, when executed by the processor circuitry/logic **204**, match a brand description string of the received consumable record with at least one label of the food knowledge graph **230**. Particularly, in one embodiment, the brand labeler **608** is configured to match the brand description string of the received consumable record with a most similar brand name label of the food knowledge graph **230**. Finally, in the case that the received consumable record is a recipe, the ingredient labeler **610** is configured to, when executed by the processor circuitry/logic **204**, match one or more ingredient description strings of the received consumable record with at least one label of the food knowledge graph **230**. Particularly, in one embodiment, the brand labeler **608** is configured to match each ingredient in the one or more ingredient description strings of the received consumable record with a most similar generic consumable name of the food knowledge graph **230**. In one embodiment, the food labeler **600** includes a label aggregator configured to aggregate or otherwise process the matched labels to provide the set of matched labels **604**.

(92) FIG. **6B** shows an exemplary process for matching a consumable record **614** to a set of labels of the food knowledge graph **230** (only partially shown). Particularly, the consumable record **614** is received by the processor circuitry/logic **204** and includes an item description field **616**, a brand description field **618**, and nutritional content data **620**. The item description field includes the item description string “Cheeze Pizza xtra cheese (L) (new menu updated **2018**)” and the brand description field includes the brand description string “the Piza hut on 11.sup.st.” As can be seen, the item description string and the brand description string include typographical errors (e.g., “Cheeze” and “Piza”), non-standard acronyms (e.g., “xtra”), in-consistent capitalization, and extraneous information (e.g. “new menu updated 2018” and “on 11.sup.st.”).

(93) In the example shown, the dish labeler **606** of the food labeler **600** has directly matched the item description string (i.e., “Cheeze Pizza xtra cheese (L) (new menu updated 2018)”) to the label “cheese pizza,” which is the most similar generic consumable name label. Similarly, the brand labeler **608** of the food labeler **600** has directly matched the brand description string (i.e., “the Piza hut on 11.sup.st.”) to the label “Pizza Hut®,” which is the most similar brand name label. The matched generic consumable name label (i.e., “cheese pizza”) and matched brand name label (i.e., “Pizza Hut®”) can be considered the direct matches of the descriptive strings of received consumable record **614**. In contrast with the item description and brand description strings, the matched generic consumable name and brand name labels are standardized and free of errors, making them more useful for performing database operations, such as search.

(94) In some embodiments, the food labeler **600** is further configured to identify additional labels of the food knowledge graph **230** that are related to the directly matched labels, as defined by the food knowledge graph **230**. In the example shown, the dish labeler **606** identifies that “cheese pizza” is a subclass of “pizza” based on the ontological relationships defined by the food knowledge graph **230**. Accordingly, the dish labeler **606** further matches the “pizza” generic consumable name label to the item description string of consumable record **614**. Similarly, the dish labeler **606** identifies that “cheese pizza” and “pizza” are in the “pie” category of consumables based on the ontological relationships defined by the food knowledge graph **230**. Accordingly, the dish labeler **606** further matches the “pie” category of consumables label to the item description string of consumable record **614**. Furthermore, the dish labeler **606** identifies that “cheese pizza” has the substitute “cheese ravioli” based on the substitute relationships defined by the food knowledge graph **230**. Accordingly, the dish labeler **606** further matches the “cheese ravioli” generic consumable name label to the item description string of consumable record **614**.

(95) The dish labeler **606** also identifies that “cheese pizza” has the ingredients “mozzarella cheese,” “pizza sauce,” and “pizza dough” based on the component relationships defined by the food knowledge graph **230**. Similarly, the dish labeler **606** identifies that “pizza sauce” has the ingredients “tomato” and “salt” and that “pizza dough” has the ingredients “flour,” “sugar,” “yeast,” “olive oil,” and “salt” based on the component relationships defined by the food

knowledge graph **230**. Accordingly, the dish labeler **606** further matches the “mozzarella cheese,” “pizza sauce,” “pizza dough,” “tomato,” “flour,” “sugar,” “yeast,” “olive oil,” and “salt” generic consumable name labels to the item description string of consumable record **614**.

(96) The dish labeler **606** further identifies that “cheese pizza” contains the allergens “lactose” and “gluten” based on the allergen relationships defined by the food knowledge graph **230**.

Accordingly, the dish labeler **606** further matches the “lactose” and “gluten” allergen labels to the item description string of consumable record **614**. Similarly, the dish labeler **606** identifies that “cheese pizza” is compliant with the “kosher” and “vegetarian” dietary restrictions based on the dietary restriction relationships defined by the food knowledge graph **230**. Accordingly, the dish labeler **606** further matches the “kosher” and “vegetarian” dietary restriction labels to the item description string of consumable record **614**. Furthermore, the dish labeler **606** identifies that “cheese pizza” has flavors “savory” and “salty” based on the flavor relationships defined by the food knowledge graph **230**. Accordingly, the dish labeler **606** further matches the “savory” and “salty” flavor labels to the item description string of consumable record **614**. Finally, the dish labeler **606** identifies that “cheese pizza” is from the cuisines “Italian” and “American” based on the cuisine relationships defined by the food knowledge graph **230**. Accordingly, the dish labeler **606** further matches the “Italian” and “American” cuisine labels to the item description string of consumable record **614**.

(97) Also in the example shown, the brand labeler **608** identifies that “Pizza Hut®” is in the “restaurant” category of brand names based on the ontological relationships defined by the food knowledge graph **230**. Accordingly, the brand labeler **608** further matches the “restaurant” category of brand names label to the brand description string of consumable record **614**.

(98) The method **500** continues with a step of at least one information field of the data record to associate the data record with the at least one first label to which the descriptive string was matched (block **540**). Particularly, the processor circuitry/logic **204** is configured to update and/or modify at least one information field of the received consumable record to associate the consumable record with the set of matched labels **604** provided by the labeler **600**. In one embodiment, the consumable record simply includes a labels field and the processor circuitry/logic **204** is configured to add the matched labels or references thereto into the labels field. However, in some embodiments, the consumable record includes several fields corresponding to the various types of labels and the processor circuitry/logic **204** is configured to add the matched labels or references thereto into the appropriate corresponding fields of the consumable record.

(99) FIG. **6C** shows an exemplary supplemented consumable record **614'** which has been updated based on the labels to which the consumable record **614** of FIG. **6B** was matched. Particularly, a generic_name field **622** of the supplemented consumable record **614'** is updated to associate the record with the “cheese pizza” and “pizza” generic consumable name labels. Similarly, a consumable_category **624** field of the supplemented consumable record **614'** is updated to associate the record with the “pie” category of consumables label. A brand_name field **626** of the supplemented consumable record **614'** is updated to associate the record with the “Pizza Hut®” brand name label. A brand_category field **628** of the supplemented consumable record **614'** is updated to associate the record with the “restaurant” category of brand names label. An ingredients field **630** of the supplemented consumable record **614'** is updated to associate the record with the “mozzarella cheese,” “pizza sauce,” “pizza dough,” “tomato,” “flour,” “sugar,” “yeast,” “olive oil,” and “salt” generic consumable name labels. An allergen field **632** of the supplemented consumable record **614'** is updated to associate the record with the “lactose” and “gluten” allergen labels. A substitutes field **634** of the supplemented consumable record **614'** is updated to associate the record with the “cheese ravioli” generic consumable name label. A flavors field **636** of the supplemented consumable record **614'** is updated to associate the record with the “savory” and “salty” flavor labels. A cuisine field **638** of the supplemented consumable record **614'** is updated to associate the record with the “Italian” and “American” cuisine labels. Finally, a dietary_compliance field **640** of

the supplemented consumable record **614'** is updated to associate the record with the “kosher” and “vegetarian” dietary restriction labels.

(100) As discussed in more detail below, in many embodiments, the processor circuitry/logic **204** of the system server **200** receives some kind of request for data consumable from a health tracking device **110**, such as search request or a recommendations request. The processor circuitry/logic **204** is configured to matching the request for consumable records to one or more relevant descriptive labels of the food knowledge graph **230**. Based on the matched descriptive labels, the processor circuitry/logic **204** is configured to provide one or more consumable records to the health tracking device **110** that are associated with the matched descriptive labels.

(101) FIG. 7 shows a method **700** of operating the health tracking system **100** to provide improved search results using the food knowledge graph **230**. Particularly, the method **700** utilizes the food knowledge graph **230**, the additional information and/or metadata generated therewith, to provide search results based on a search string. The method **700** improves the functioning of the system server **200** by enabling the processor circuitry/logic **204** to generate a list of consumable records based on search string that are more relevant than would be provided based only on conventional text-comparison based searching.

(102) FIG. 8A shows an exemplary graphical user interface displayed on the smartphone **110A**, in which a search of the consumable records **224** was performed in a conventional manner only on the basis of the user-generated item description and brand description strings of the consumable records. As can be seen, a user has entered the search string “mac n cheese” into a search window **810** of the graphical user interface. However, the search results **820** displayed on the graphical user interface include several prominently ranked consumable records that are duplicative, of low relevance, and of low quality. For example, in the search results **820**, the second result having the item description “Generic,” the third result having the item description “Homemade Macaroni and Cheese,” and the fifth result having the item description “Mac N' Cheese,” can be considered mostly duplicative of one another because they are all generic records. Additionally, the sixth result having the item description “Macaroni and Cheese Pie,” and the eighth result having the item description “Ham and Cheese Macaroni” are likely of low relevance to the search string “mac n cheese” because a user actually searching for these records would have likely used a more specific search term including the word “pie” or “ham,” respectively. Furthermore, the second result can be considered low quality because the item and brand descriptions are reversed (i.e. the item description is “Generic” and brand description is “Macaroni and Cheese”). Similarly, the fourth result can also be considered low quality because it is ambiguous what the consumable item actually is (i.e., the item description is “Cheese” and the brand description is “Macaroni”).

(103) Returning to FIG. 7, the method **700** uses the food knowledge graph **230** to eliminate many of the issues with a conventional text-comparison based search. Particularly, the method **700** begins with a step of storing a plurality of labels describing consumable items and a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items (block **710**). Particularly, with respect to the embodiments disclosed in detail herein, the processor circuitry/logic **204** of the system server **200** is configured to operate the database **220** to store the food knowledge graph **230**. The food knowledge graph **230** comprises a plurality of descriptive labels describing consumable items and a plurality of relationships between pairs or groups of labels in the plurality of descriptive labels. In one embodiment, the descriptive labels of the food knowledge graph **230** at least include generic consumable names, but may include any other type of descriptive label discussed above with respect to FIGS. 4A-4E or any other useful descriptive labels not discussed above. Furthermore, the plurality of relationships of the food knowledge graph **230** may include any of the relationship types discussed above with respect to FIGS. 4A-4E, as well as any other useful relationship type not discussed above. In some embodiments, the processor circuitry/logic **204** is configured to operate the database **220** to store the food knowledge graph **230** in the form of one or more lists or

tables that define a data tree or data web in which the descriptive labels comprise the nodes of the tree or web and the relationships comprise the connections between the nodes of the tree or web (for example, as illustrated in FIGS. 4A-4E). In one embodiment, the processor circuitry/logic **204** is configured to operate the database **220** to store the food knowledge graph **230** in the form of a reduced number of self-referential lists or tables, possible a single self-referential list or table, that define all of the labels and relationships that form the data tree or data web. In some embodiments, the processor circuitry/logic **204** is configured to operate the database **220** to store the labels and/or relationships as a list or table of triplets formed as, for example, [label_ID1, relationship_type, label_ID2] where label_ID1 is a first descriptive label or reference thereto, label_ID2 is a second descriptive label or reference thereto, and relationship_type indicates the nature of the relationship. However, the food knowledge graph can take any other suitable form.

(104) The method **700** continues with a step of storing a plurality of consumable records, each consumable record including at least a descriptive string regarding a consumable item and each consumable record being associated with at least one label in the plurality of labels (block **720**). Particularly, as discussed above, the processing circuitry/logic **204** of the server **200** is configured to maintain a consumable records database **224** in the memory **206**. Each consumable record comprises a plurality of data fields that relate to a particular consumable item, including at least one description field have a descriptive string. In some embodiments, many, if not all, of consumable records **224** are created by users of the health tracking system **100** and have at least one user-generated text description field (e.g., the item description and brand description strings, discussed elsewhere herein). In addition, each consumable record is associated with one or more descriptive labels of the food knowledge graph **230**. Particularly, in some embodiments, each consumable record includes one or more supplemental fields having references to descriptive labels of the food knowledge graph **230**. As discussed in more detail with respect to FIG. 6C, some exemplary supplemental fields having references to descriptive labels of the food knowledge graph **230** may include fields for a standardized generic consumable name, a consumable category, a standardized brand name, a brand name category, dietary substitutes, included ingredients, included allergens, flavors, a type of cuisine, dietary restriction compliance, and other miscellaneous useful descriptive labels. In one embodiment, the supplemental fields were generated according to the method **500** or some similar labeling process.

(105) The method **700** continues with a step of receiving a search string from the health tracking device of the user (block **730**). Particularly, the processing circuitry/logic **204** of the system server **200** is configured to operate the network communications module **212** to receive a search string from a health tracking device **110**, such as the smartphone **110A** (e.g., “mac n cheese” in the example of FIG. 8A). In at least one embodiment, the processor **308** of one of the health tracking device **110** is configured to execute instructions of the client-side health tracking application **316** to enable a user to enter a search string, which will be used as the search string. The processor **308** is configured to operate the transceivers **312** to transmit the search string to the system server **200**. The processing circuitry/logic **204** of the system server **200** is configured to operate the network communications module **212** to receive the search string from the health tracking device **110**.

(106) The method **700** continues with a step of matching the search string to at least one first label in the plurality of labels (block **740**). Particularly, the processor circuitry/logic **204** is configured to execute a search labeling process of the health tracking application **218** to match the search string with one or more descriptive labels of the food knowledge graph **230**. The search labeling process is essentially similar to, and possibly the same as, the food labeler **600** described above. The search labeling process is configured to, when executed by the processor circuitry/logic **204**, match the search string with one or more descriptive labels of the food knowledge graph **230** and output a one or more matched labels. In one embodiment, the search labeling process is configured to compare the text of the search string with the labels and match the search string to the most similar label(s) of the food knowledge graph **230** or to all label(s) of the food knowledge graph **230** having at least

a threshold level of similarity. In some embodiments, rather than using a rules-based text comparison, the search labeling process is configured to, when executed by the processor circuitry/logic **204**, instead use a machine learning model to match the search string to labels of the food knowledge graph **230**.

(107) In some embodiments, the search labeling process is configured to, when executed by the processor circuitry/logic **204**, match the search string only to a single most similar descriptive label, which might be a generic consumable name, a branded consumable, a brand name, a category of consumables, or another of the other types of labels described herein. For example, if the search string is “mac n cheese,” as in the example of FIG. **8A**, the processor circuitry/logic **204** might match the search string to the generic consumable name label “macaroni and cheese.” Similarly, if the search string is “fruit,” the processor circuitry/logic **204** might match the search string to the category of consumables label “fruit.” However, in some embodiments, the search labeling process is configured to, when executed by the processor circuitry/logic **204**, match the search string several labels in a similar manner as the food labeler **600**. For example, if the search string is “mac n cheese,” as in the example of FIG. **8A**, the processor circuitry/logic **204** might match the search string to the generic consumable name label “macaroni and cheese,” as well as the category of consumables label “pasta,” or other related descriptive labels of the food knowledge graph **230**. Additionally, for example, if the search string is “kraft mac n cheese,” the processor circuitry/logic **204** might match the search string to the most similar generic consumable name label “macaroni and cheese” and to the most similar brand name label “Kraft®.”

(108) The method **700** continues with steps of generating a list of consumable records including consumable records in the plurality of consumable records that are associated with the at least one first label to which the search string was matched (block **750**) and providing the list of consumable records to the health tracking device as search results (block **760**). Particularly, the processor circuitry/logic **204** is configured to generate a list of consumable records from the consumable records **224** in the database **220** that are associated with and/or have the label(s) that was matched to the search string. The processor circuitry/logic **204** is configured to operate the network communications module **212** to transmit the generated list of consumable records to the health tracking device **110**, such as the smartphone **110A**, from which the search string was received. The processor **308** of the respective health tracking device **110** is configured to present the list of consumable records to the user as search results via a search results screen of a graphical user interface on the display screen **302**.

(109) FIG. **8B** shows an exemplary graphical user interface displayed on the smartphone **110A**, in which a search of the consumable records **224** was performed by matching the search string to the food knowledge graph **230**. As in the example of FIG. **8A**, a user has entered the search string “mac n cheese” into a search window **810** of the graphical user interface. Improved search results **830** are displayed on the graphical user interface. The search results **830** only comprise consumable records that are associated with the generic consumable name label “macaroni and cheese” to which the search string was matched. Accordingly, the search results **830** only include consumable records that are relevant to the search string, and implicitly exclude records which are only superficially relevant due to containing similar words in their text descriptions (e.g., consumable records for macaroni noodles or consumable records for cheese).

(110) In some embodiments, the ranking of the list of consumable records may be in accordance with the comparative popularity of each of the records, or by some other ranking method. Particularly, in some embodiments, each consumable record in the database **220** may be assigned a popularity score based on how frequently the consumable record is logged by users of the health tracking system **100**. In at least one embodiment, the consumable records associated with the matched descriptive labels are ranked in the list according to their respective popularity score. In other embodiments, other scores may be assigned to consumable records and also used to rank the list of records, such as an accuracy or quality score. In addition some consumable records in the

database **220** that are of high quality and accuracy may be automatically or manually marked as “verified” records by an administrator of the database **220** (indicated by a check mark **832** in the graphical user interface of FIG. **8B**). These verified records may be automatically ranked higher in the list of consumable records that are provided as search results.

(111) In some embodiments, each generic consumable name label and/or branded consumable name in the food knowledge graph **230** is associated with a particular consumable record that is considered to be the canonical consumable record or official consumable record for the respective generic consumable item and/or branded consumable item. In some embodiments, each particular combination a generic consumable name label and a brand name label may be associated with a canonical branded consumable record. In one embodiment, if the search string is matched to a generic consumable name label, but is not matched to any brand name labels or branded consumable name labels, then the processor circuitry/logic **204** is configured to provide the canonical generic consumable record at the top of the generated list of consumable records. In the example of FIG. **8B**, the search string was matched to the “macaroni and cheese” generic consumable name label, but is not matched to any brand name labels or branded consumable name labels. Accordingly, a canonical record **834** having item description “Macaroni and Cheese” and brand description “Generic” is automatically provided at the top of the search results **830**. Similarly, if a search string is matched to a generic consumable name label and a brand name label, the processor circuitry/logic **204** is configured to provide the canonical consumable record corresponding to that particular combination a generic consumable name label and a brand name label.

(112) In some embodiments, if the search string is matched to a generic consumable name label, particularly if the search string is also not matched to any particular brand name labels or branded consumable name labels, then the processor circuitry/logic **204** is also configured to provide the most popular branded consumable records in the database **220** that are associated with matched generic consumable name label, as determined based on their respective popularity scores. For example, in the example of FIG. **8B**, the search string was matched to the “macaroni and cheese” generic consumable name label, but is not matched to any brand name labels or branded consumable name labels. Accordingly, as shown in FIG. **8B**, popular consumable records from the brands “Velveta,” “Kraft,” “Morrison,” “Panera,” “Annie's Homegown,” “Amy's,” and “Bojangles” are automatically provided in the search results **830**, below the canonical generic consumable record **834**.

(113) In some embodiments, the processor circuitry/logic **204** is configured to operate the database **220** to store a list of sponsored consumable records and/or to mark a certain subset of the consumable records **224** as being sponsored. In one embodiment, the list of sponsored consumable records is managed by an administrator of the health tracking system **100**. After matching the search string to a descriptive label of the food knowledge graph **230**, the processor circuitry/logic **204** is configured to identify whether any of the sponsored consumable records are associated with the matched descriptive label. If so, the processor circuitry/logic **204** is configured to automatically include the sponsored consumable records associated with the matched descriptive label in the generated list. For example, as shown in FIG. **8B**, a sponsored consumable record **836** was associated with the generic consumable name label “macaroni and cheese” and was automatically included in the search results **830**.

(114) In some embodiments, after generating the list of consumable records to be presented as search results to the user, the processor circuitry/logic **204** is configured to compare the labels with which each of the consumable records in list is associated. If two consumable records in the list are associated with exactly the same set of labels, then the processor circuitry/logic **204** is configured to exclude one of them from the list. For example, if two records are associated with the same generic consumable name label and the same brand name label, there is an implication that the two records are both for the same brand of the same consumable item, and thus are duplicative of one

another. In other embodiments, other rules for identifying and excluding duplicative consumable records based on the associated labels can be utilized.

(115) The method **700** also enables the user to search in a more diverse manner than could be accomplished with only a conventional text-comparison based search. Particularly, if a user enters the search string “fruit,” a text-comparison based search would simply yield an assortment of consumable records that happen to have the word “fruit” in their item description or brand description (e.g. “Fruit Roll Up®” or “Mixed Fruit Cup”). However, in the method **700**, the search string “fruit” would be matched to the category of consumables label “fruit.” In some embodiments, when a search string matches best to a non-consumable name label, such as a category label, a brand name label, an allergen label, a dietary restriction label, or a flavor label, the processor circuitry/logic **204** is configured to generate the list of consumable records with the most popular consumable records associated with the non-consumable name label. In one embodiment, the list is generated with the most popular generic consumable records associated with the non-consumable name label. For example, if the search string matches to the category of consumables label “fruit,” then the search results may include the most popular generic consumable records associated with the category of consumables label “fruit” (e.g., generic records for an apple, orange, watermelon, lemon, strawberry, etc.). Similarly, if the search string matches to the cuisine label “Italian,” then the search results may include the most popular consumable records associated with the cuisine label “Italian,” (e.g., records for various types of pasta, pizza, etc.).

(116) FIG. **9** shows a method **900** of operating the health tracking system **100** to provide improved recommendations using the food knowledge graph **230**. Particularly, the method **900** utilizes the food knowledge graph **230**, the additional information and/or metadata generated therewith, to provide recommendations based on a selection of one or more consumable records by a user. The method **900** improves the functioning of the system server **200** by enabling the processor circuitry/logic **204** to generate a list of recommended consumable records that are more relevant than would be provided based only on text-comparison based determinations of recommended consumable records.

(117) The method **900** begins with a step of storing a plurality of labels describing consumable items and a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items (block **910**). Particularly, with respect to the embodiments disclosed in detail herein, the processor circuitry/logic **204** of the system server **200** is configured to operate the database **220** to store the food knowledge graph **230**. The food knowledge graph **230** comprises a plurality of descriptive labels describing consumable items and a plurality of relationships between pairs or groups of labels in the plurality of descriptive labels. In one embodiment, the descriptive labels of the food knowledge graph **230** at least include generic consumable names, but may include any other type of descriptive label discussed above with respect to FIGS. **4A-4E** or any other useful descriptive labels not discussed above. Furthermore, the plurality of relationships of the food knowledge graph **230** may include any of the relationship types discussed above with respect to FIGS. **4A-4E**, as well as any other useful relationship type not discussed above. In some embodiments, the processor circuitry/logic **204** is configured to operate the database **220** to store the food knowledge graph **230** in the form of one or more lists or tables that define a data tree or data web in which the descriptive labels comprise the nodes of the tree or web and the relationships comprise the connections between the nodes of the tree or web (for example, as illustrated in FIGS. **4A-4E**). In one embodiment, the processor circuitry/logic **204** is configured to operate the database **220** to store the food knowledge graph **230** in the form of a reduced number of self-referential lists or tables, possible a single self-referential list or table, that define all of the labels and relationships that form the data tree or data web. In some embodiments, the processor circuitry/logic **204** is configured to operate the database **220** to store the labels and/or relationships as a list or table of triplets formed as, for example, [label_ID1, relationship_type, label_ID2] where label_ID1 is a first descriptive

label or reference thereto, label_ID2 is a second descriptive label or reference thereto, and relationship_type indicates the nature of the relationship. However, the food knowledge graph can take any other suitable form.

(118) The method **900** continues with a step of storing a plurality of consumable records, each consumable record including at least a descriptive string regarding a consumable item and each consumable record being associated with at least one label in the plurality of labels (block **920**). Particularly, as discussed above, the processing circuitry/logic **204** of the server **200** is configured to maintain a consumable records database **224** in the memory **206**. Each consumable record comprises a plurality of data fields that relate to a particular consumable item, including at least one description field have a descriptive string. In some embodiments, many, if not all, of consumable records **224** are created by users of the health tracking system **100** and have at least one user-generated text description field (e.g., the item description and brand description strings, discussed elsewhere herein). In addition, each consumable record is associated with one or more descriptive labels of the food knowledge graph **230**. Particularly, in some embodiments, each consumable record includes one or more supplemental fields having references to descriptive labels of the food knowledge graph **230**. As discussed in more detail with respect to FIG. **6C**, some exemplary supplemental fields having references to descriptive labels of the food knowledge graph **230** may include fields for a standardized generic consumable name, a consumable category, a standardized brand name, a brand name category, dietary substitutes, included ingredients, included allergens, flavors, a type of cuisine, dietary restriction compliance, and other miscellaneous useful descriptive labels. In one embodiment, the supplemental fields were generated according to the method **500** or some similar labeling process.

(119) The method **900** continues with a step of receiving from a user, at least one consumable item selection corresponding to a consumable record in the plurality of consumable records in the database (block **930**). Particularly, the processing circuitry/logic **204** of the system server **200** is configured to operate the network communications module **212** to consumable item selection corresponding to a consumable record in the database from a health tracking device **110**, such as the smartphone **110A** (e.g., “mac n cheese” in the example of FIG. **8A**). In at least one embodiment, the processor **308** of one of the health tracking device **110** is configured to execute instructions of the client-side health tracking application **316** to enable a user to view the details of the various consumable records **224** in the database **220**, as well as add individual consumable records to a food log or diary of the user. As used herein, a “consumable item selection” refers to an interaction of the user with one or more particular consumable records via a respective health tracking device **110**, such as pressing or clicking on a consumable record in a list of consumable records to view the details of an individual consumable record, selecting one or more particular consumable records from a list of consumable records to log them to a food log or diary, or any other request from the health tracking device **110** regarding one or more particular consumable records in the databases **220**. The processor **308** is configured to operate the transceivers **312** to transmit, to the system server **200**, a request or message indicating a selection of a consumable record. The processing circuitry/logic **204** of the system server **200** is configured to operate the network communications module **212** to receive the request or message indicating a selection of a consumable record from the health tracking device **110**. If the request or message indicates that the user would like to view the details of a particular consumable record, the processing circuitry/logic **204** may be configured to operate the network communications module **212** to transmit the details of the particular consumable record. Likewise, if the request or message indicates that the user would like to log the particular consumable record to a food log or diary, the processing circuitry/logic **204** may be configured to update the food log or diary based on the selected consumable record.

(120) In at least one embodiment, the processor **308** of one of the health tracking device **110** is configured to execute instructions of the client-side health tracking application **316** to enable a user

to request recommendations of consumable records from the system server **200**. The processor **308** is configured to operate the transceivers **312** to transmit, to the system server **200**, a recommendations request indicating that the user wishes to receive recommendations. In some embodiments, the recommendations request may indicate that the user wishes to receive recommendations that are similar to, alternatives to, or substitutes of a particular selected consumable record. In some embodiments, the recommendations request may indicate that the user wishes to receive recommendations based on the types of consumables that they have previously added to their food log or diary. The processing circuitry/logic **204** of the system server **200** is configured to operate the network communications module **212** to receive the recommendations request from the health tracking device **110**.

(121) The method **900** continues with steps of identifying at least one first label in the plurality of labels that is associated with the consumable record in the plurality of consumable records corresponding to the at least one consumable item selection (block **940**) and providing a list of consumable records to the health tracking device as recommended consumable items for the user based on the at least one first label (block **950**). Particularly, the processing circuitry/logic **204** of the system server **200** is configured to identify which descriptive labels of the food knowledge graph **230** are associated with the consumable record(s) corresponding to the received consumable item selection, in response to receiving the consumable item selection and/or in response to receiving a recommendations request. Next, the processor circuitry/logic **204** is configured to generate a list of consumable records from the consumable records **224** in the database **220** based on the identified set of labels associated with the selected consumable record(s). The processor circuitry/logic **204** is configured to operate the network communications module **212** to transmit the generated list of consumable records to the health tracking device **110**, such as the smartphone **110A**, from which the consumable item selection or recommendations request was received. The processor **308** of the respective health tracking device **110** is configured to present the list of consumable records to the user as recommendations via a recommendations screen of a graphical user interface on the display screen **302**.

(122) In at least one embodiment, in response to a recommendations request with respect to a particular consumable item selection, the processor circuitry/logic **204** is configured to generate a list of consumable records that are similar to, substitutes of, and/or alternatives the selected consumable record based on the identified set of labels associated with the particular selected consumable record. FIG. **10A** shows an exemplary graphical user interface displayed on the smartphone **110A**, in which recommendations of similar consumable records are provided using the food knowledge graph **230**. Particularly, a user selected a consumable record with the item description “Fiber One® Oats & Chocolate Bar” and requested recommendation of consumable records from the databases **220** that are similar to the selected consumable record by pressing a similar option **1012**. The graphical user interface shows a title **1010** indicating the nature of the recommendations (e.g., “Similar to “Fiber One® Oats & Chocolate Bar”) and a list **1020** of recommended consumable records. In the example, the processor circuitry/logic **204** identified that the selected consumable was associated with, for example, the generic consumable name “granola bar,” the category of consumables “on-the-go snacks,” and the ingredient names “oats,” “granola,” and “chocolate.” As can be seen, the list **1020** includes consumable records for various other on-the-go snacks that are similar to the “Fiber One® Oats & Chocolate Bar” consumable record.

(123) To generate a list of recommended consumable records that are similar to a particular consumable record, the processor circuitry/logic **204** is configured to identify the descriptive labels associated with the particular consumable record, and include consumable records in the list that are associated with a similar set of descriptive labels. In one embodiment, the processor circuitry/logic **204** is configured to include consumable records in the list only if they are associated with a minimum threshold amount of the same descriptive labels as the particular consumable record (e.g., at least 75% overlap in the associated labels). In one embodiment, the

processor circuitry/logic **204** is configured to include consumable records in the list only if they are associated with a same category of consumables label as the particular consumable record. In one embodiment, the processor circuitry/logic **204** is configured to exclude consumable records that are associated with exactly the same set of labels as the selected consumable record (i.e., 100% overlap in the associated labels), to avoid recommending duplicative records or minor variants of the selected consumable record.

(124) In some embodiments, a recommendation request may include an option or selection to show consumable records which are dietary substitutes to a selected consumable record. Particularly, the graphical user interface of FIG. **10A** includes a substitutes option **1014**, which can be pressed by the user. In response to the user pressing the substitutes option **1014**, the processor **308** of the health tracking device **110** is configured to transmit a recommendations request indicating that the user wishes to receive recommendations of substitutes. When generating the list of recommended consumable records, the processor circuitry/logic **204** is configured to only include consumable records which are dietary substitutes of the selected consumable record, as defined by food knowledge graph **230**. Depending on how the food knowledge graph **230** defines substitutes, the generated list of recommended substitute consumable records may require much more narrow similarity, as opposed to the recommendation of merely “similar” consumable records.

(125) In some embodiments, a recommendation request may include an option or selection to show consumable records which are healthier alternatives to a selected consumable record. Particularly, the graphical user interface of FIG. **10A** includes a healthier option **1016**, which can be pressed by the user. In response to the user pressing the healthier option **1016**, the processor **308** of the health tracking device **110** is configured to transmit a recommendations request indicating that the user wishes to receive recommendations of healthier alternatives. In some embodiments, at least some of the consumable records **224** in the database **220** each have a health score associated therewith, which is determined based on the nutritional information of the respective consumable record. When generating the list of recommended consumable records, the processor circuitry/logic **204** is configured to only include consumable records which are associated with a similar set of descriptive labels, but have a higher health score. In other embodiments, the processor circuitry/logic **204** is configured to populate the list of recommended consumable records based on some other method of evaluating the healthiness of the consumable records.

(126) In some embodiments, a recommendation request may include options or selections for filtering the list of recommended consumable records, such as an option or selection to filter out any consumable records associated with a particular allergen. Particularly, the graphical user interface of FIG. **10A** includes a filters option **1018**, which can be pressed by the user to provide a menu of filter selections that can be selected by the user to filter the recommendations in various ways. In response to the user pressing the filters option **1018** and selecting a filter, the processor **308** of the health tracking device **110** is configured to transmit a recommendations request indicating that the user wishes to filter the recommendations of consumable records according to the selected filter. In one example, in response to the recommendations request including an allergen filter selection, the processor circuitry/logic **204** is configured to exclude consumable records from the list of recommended consumable records that are associated with the allergen, based on the food knowledge graph **230**.

(127) In some embodiments, a recommendation request may include an option or selection to show consumable records having the same brand as a selected consumable record. Particularly, when generating the list of recommended consumable records, the processor circuitry/logic **204** is configured to only include consumable records which are associated with the same brand label as the selected consumable record.

(128) In at least one embodiment, in response to a recommendations request indicating that a user wishes to receive recommendations of consumable records based on the types of consumables that they have previously added to their food log or diary, the processor circuitry/logic **204** is

configured to generate a list of consumable records that are similar to, substitutes of, and/or alternatives to the selected consumable records previously added to their food log or diary. FIG. **10B** shows an exemplary graphical user interface displayed on the smartphone **110A**, in which user-specific recommendations of consumables are provided using the food knowledge graph. The graphical user interface shows a title **1030** indicating the nature of the recommendations (e.g., 'Recommended for you'). A user has selected that he or she would like to receive recommendations of consumable records from the databases **220** based on the types of consumables that they have previously added to their food log or diary by pressing a recommended for me option **1032**. In response to the selection, the processor **308** of the health tracking device **110** is configured to transmit a recommendations request indicating that the user wishes to receive recommendations of consumable records based on the types of consumables that they have previously added to their food log or diary.

(129) In at least one embodiment, the processor circuitry/logic **204** is configured to generate a user dietary profile based on all of the consumable records that the user has previously added to their food log or diary, or based on the consumable records that the user has previously added to their food log or diary within a predetermined time period (e.g. the last six months). The processor circuitry/logic **204** is configured to identify, for each consumable record added to their food log or diary, which descriptive labels of the food knowledge graph **230** are associated therewith. Next, the processor circuitry/logic **204** is configured to calculate, for each descriptive label associated with the previously logged consumable records, a frequency at which the user has logged consumable records associated the respective label. These calculated frequencies and/or percentages form a user profile for the user from which recommendations of further consumable records can be made.

(130) Returning to the example of FIG. **10B**, the graphical user interface shows a list **1040** of recommended consumable records that are similar to consumable records that the user previously added to their food log or diary. Based the labels associated with the previously logged consumable records, the processor circuitry/logic **204** calculated that, for example, 50% of the consumable records logged by the user are associated with the "Italian" cuisine label, 15% of the consumable records logged by the user are associated with the "pizza" generic consumable name label, 20% of the consumable records logged by the user are associated with the "pasta" generic consumable name label, etc. As a result, the list **1040** of recommended consumable records includes several consumable records having similar associated descriptive labels to that of the user's dietary profile.

(131) To generate a list of recommended consumable records based on those that the user has previously added to their food log or diary, the processor circuitry/logic **204** is configured to identify a subset of the descriptive labels that are most frequently associated with consumable records that the user has previously added to their food log or diary. When generating the list of recommended consumable records, the processor circuitry/logic **204** is configured to include consumable records that are associated with the most frequent labels. In one embodiment, the processor circuitry/logic **204** is configured to include consumable records in the list only if they are associated with a minimum threshold amount of the identified subset of the most frequent labels (e.g., at least 10% overlap with the most frequent labels).

(132) In some embodiments, a recommendation request may include an option or selection to show consumable records which are healthier alternatives to a selected consumable record. Particularly, the graphical user interface of FIG. **10B** includes a healthier option **1034**, which can be pressed by the user. In response to the user pressing the healthier option **1034**, the processor **308** of the health tracking device **110** is configured to transmit a recommendations request indicating that the user wishes to receive recommendations of healthier alternatives. In some embodiments, at least some of the consumable records **224** in the database **220** each have a health score associated therewith, which is determined based on the nutritional information of the respective consumable record. When generating the list of recommended consumable records, the processor circuitry/logic **204** is configured to only include consumable records which are associated with a similar set of

descriptive labels as the most frequent labels, but have a higher health score. In other embodiments, the processor circuitry/logic **204** is configured to populate the list of recommended consumable records based on some other method of evaluating the healthiness of the consumable records.

(133) In some embodiments, a recommendation request may include options or selections for filtering the list of recommended consumable records, such as an option or selection to filter out any consumable records associated with a particular allergen.

(134) Particularly, the graphical user interface of FIG. **10B** includes a filters option **1036**, which can be pressed by the user to provide a menu of filter selections that can be selected by the user to filter the recommendations in various ways. In response to the user pressing the filters option **1036** and selecting a filter, the processor **308** of the health tracking device **110** is configured to transmit a recommendations request indicating that the user wishes to filter the recommendations of consumable records according to the selected filter. In one example, in response to the recommendations request including an allergen filter selection, the processor circuitry/logic **204** is configured to exclude consumable records from the list of recommended consumable records that are associated with the allergen, based on the food knowledge graph **230**.

(135) The herein described applications utilizing food knowledge graph **230** (e.g., the health tracking program **218** and/or health tracking application **316**) improve the functioning of the processing circuitry/logic **204** and/or the processor **308**, respectively or in combination by enabling it/them to provide more informative, efficient, and diverse user interactions with the consumable records **224** in the database **220**. Particularly, the method **500** improves the functioning of the system server **200** by enabling the processor circuitry/logic **204** to utilize the structured knowledge defined by the food knowledge graph **230** to provided additional information and/or metadata which is stored in or referenced by consumable records, in addition to the user-generated information of the consumable records. Additionally, the method **700** improves the functioning of the system server **200** by enabling the processor circuitry/logic **204** to generate a list of consumable records based on search string that are more relevant than would be provided based only on text-comparison based searching. The method **900** improves the functioning of the system server **200** by enabling the processor circuitry/logic **204** to generate a list of recommended consumable records that are more relevant than would be provided based only on text-comparison based determinations of recommended consumable records. Finally, devices that are able to utilize the food knowledge graph **230** can operate more efficiently to respond to various other types of requests for consumable records from the database **220**.

(136) It will be appreciated that the various ones of the foregoing aspects of the present disclosure, or any parts or functions thereof, may be implemented using hardware, software, firmware, tangible, and non-transitory computer readable or computer usable storage media having instructions stored thereon, or a combination thereof, and may be implemented in one or more computer systems.

(137) Particularly, in some embodiments, a permanent copy of the programming instructions for individual ones of the aforementioned applications utilizing the food knowledge graph **230** (e.g., the health tracking program **218** and/or health tracking application **316**) may be placed into permanent storage devices (such as e.g., the memory **206** and/or the memory **310**) during manufacture thereof, or in the field, through e.g., a distribution medium (not shown), such as a compact disc (CD), or through communication interface **212**, **304** from a distribution server (such as the server **200** and/or another distribution server). That is, one or more distribution media having an implementation of the agent program may be employed to distribute the agent and program various computing devices.

(138) The foregoing detailed description of one or more exemplary embodiments of the health tracking system **100** has been presented herein by way of example only and not limitation. It will be recognized that there are advantages to certain individual features and functions described herein that may be obtained without incorporating other features and functions described herein.

Moreover, it will be recognized that various alternatives, modifications, variations, or improvements of the above-disclosed exemplary embodiments and other features and functions, or alternatives thereof, may be desirably combined into many other different embodiments, systems or applications. Presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the appended claims. Therefore, the spirit and scope of any appended claims should not be limited to the description of the exemplary embodiments contained herein.

Claims

1. A method of operating a health tracking system, the method comprising: storing in a database, a plurality of data records wherein each data record includes at least one descriptive string regarding one of a plurality of consumable items; storing in a memory, a food knowledge data web comprised of a plurality of nodes and a plurality of connections between the nodes, wherein the nodes of the food knowledge data web are defined by a plurality of labels describing consumable items and the connections of the food knowledge data web are defined by a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items, and wherein, for each pair of labels, one food relationship is defined between a first label and a second label of said pair of labels, said one food relationship identifying one of (i) the second label as an allergen of the first label, (ii) the second label as a substitute for the first label, and (iii) the second label as a flavor of the first label; receiving a request for a consumable item from a health tracking device; matching the request to the at least one first label or the at least one second label of one of said pair of labels in the food knowledge data web by implementing a machine learning model in order to match at least one descriptive string in the request to the at least one first label or the at least one second label of the one of said pair of labels in the food knowledge data web; in response to the request being matched to the at least one first label or the at least one second label, identifying a data record from one of the plurality of data records by matching the matched at least one first or second label to a data record from the plurality of data records in the database; transmitting the identified data record to the health tracking device; displaying a menu on the health tracking device allowing the user to select a first option for recommended similar items, a second option for recommended substitutes, and a third option for recommended healthier items in lieu of the identified data record; when the first option is selected at the health tracking device, transmitting to the health tracking device by wireless transmission a first set of one or more additional data records in the database that are associated with a first minimum threshold amount of same descriptive labels as the identified data record in the database, and immediately displaying the first set on the health tracking device following receipt of the selection of the first option; when the second option is selected at the health tracking device, transmitting to the health tracking device by wireless transmission a second set of one or more additional data records in the database that are each either (i) represented by one of the nodes in the food knowledge data web that is connected to another of the nodes in the food knowledge data web representing the identified data record by one of the connections of the food knowledge data web indicating a substitute relationship between the additional data record and the identified data record, or (ii) associated with a second minimum threshold amount of the same descriptive labels as the identified data record, wherein the second minimum threshold is greater than the first minimum threshold, and immediately displaying the second set on the health tracking device following receipt of the selection of the second option; and when the third option is selected at the health tracking device, transmitting to the health tracking device by wireless transmission a third set of one or more additional data records in the database that are associated with a third minimum threshold amount of the same descriptive labels as the identified data record in the database and also have a health score in the database that is greater than a health score of the

identified data record, and immediately displaying the third set on the health tracking device following receipt of the selection of the third option.

2. The method of claim 1 further comprising: receiving a data record comprising at least one descriptive string regarding a first consumable item from the health tracking device; and including the received data record within the database.

3. A non-transitory computer-readable medium which, when executed on one or more processors of a health tracking system, cause the one or more processors to: store in a database, a plurality of data records wherein each data record includes at least one descriptive string regarding one of a plurality of consumable items; store in a memory, a food knowledge data web comprised of a plurality of nodes and a plurality of connections between the nodes, wherein the nodes of the food knowledge data web are defined by a plurality of labels describing consumable items and the connections of the food knowledge data web are defined by a plurality of relationships between pairs of labels in the plurality of labels, at least some of the labels in the plurality of labels being generic names for consumable items, and wherein, for each pair of labels, one food relationship is defined between a first label and a second label of said pair of labels; receive a request for a consumable item from a health tracking device; match the request to the at least one first label or the at least one second label of one of said pair of labels in the food knowledge data web by implementing a machine learning model in order to match at least one descriptive string in the request to the at least one first label or the at least one second label of the one of said pair of labels in the food knowledge data web; in response to the request being matched to the at least one first label or the at least one second label, identify a data record from one of the plurality of data records by matching the matched at least one first or second label to a data record from the plurality of data records in the database; transmit the identified data record to the health tracking device; display a menu on the health tracking device allowing the user to select a first option for recommended similar items, a second option for recommended substitutes, and a third option for recommended healthier items in lieu of the identified data record; when the first option is selected at the health tracking device, transmit to the health tracking device by wireless transmission a first set of one or more additional data records in the database that are associated with a first minimum threshold amount of same descriptive labels as the identified data record in the database, and immediately displaying the first set on the health tracking device following receipt of the selection of the first option; when the second option is selected at the health tracking device, transmit to the health tracking device by wireless transmission a second set of one or more additional data records in the database that are each either (i) represented by one of the nodes in the food knowledge data web that is connected to another of the nodes in the food knowledge data web representing the identified data record by one of the connections of the food knowledge data web indicating a substitute relationship between the additional data record and the identified data record, or (ii) associated with a second minimum threshold amount of the same descriptive labels as the identified data record, wherein the second minimum threshold is greater than the first minimum threshold, and immediately displaying the second set on the health tracking device following receipt of the selection of the second option; and when the third option is selected at the health tracking device, transmit to the health tracking device by wireless transmission a third set of one or more additional data records in the database that are associated with a third minimum threshold amount of the same descriptive labels as the identified data record in the database and also have a health score in the database that is greater than a health score of the identified data record, and immediately displaying the third set on the health tracking device following receipt of the selection of the third option.

4. The non-transitory computer-readable medium of claim 3 wherein said one food relationship identifies one of (i) the second label as an allergen of the first label, (ii) the second label as a substitute for the first label, and (iii) the second label as a flavor of the first label.

5. The non-transitory computer-readable medium of claim 3 wherein the one or more processors are further configured to: receive a data record comprising at least one descriptive string regarding

a first consumable item from the health tracking device; and include the received data record within the database.
